

Assignment 2
Computer Programming with C
MCA102
Topic: C Functions

1) Write a C program to add, subtract, multiply and divide two integers using a user-defined type function with a return type.

Input :-

```
#include <stdio.h>

int add(int a, int b) {
    return a + b;
}

int subtract(int a, int b) {
    return a - b;
}

int multiply(int a, int b) {
    return a * b;
}

int divide(int a, int b) {
    if (b == 0) {
        printf("Error: Division by zero!\n");
        return 0;
    }
}
```

```
        return a / b;
    }

int main() {
    int num1, num2, result;

    char operator;

    printf("Enter two integers: ");

    scanf("%d %d", &num1, &num2);

    printf("Enter an operator (+, -, *, /): ");

    scanf(" %c", &operator);

    switch (operator) {
        case '+':

            result = add(num1, num2);

            printf("%d + %d = %d\n", num1, num2, result);

            break;

        case '-':

            result = subtract(num1, num2);

            printf("%d - %d = %d\n", num1, num2, result);

            break;

        case '*':

            result = multiply(num1, num2);

            printf("%d * %d = %d\n", num1, num2, result);

            break;

        case '/':

            result = divide(num1, num2);

            printf("%d / %d = %d\n", num1, num2, result);
```

```
        break;

    default:

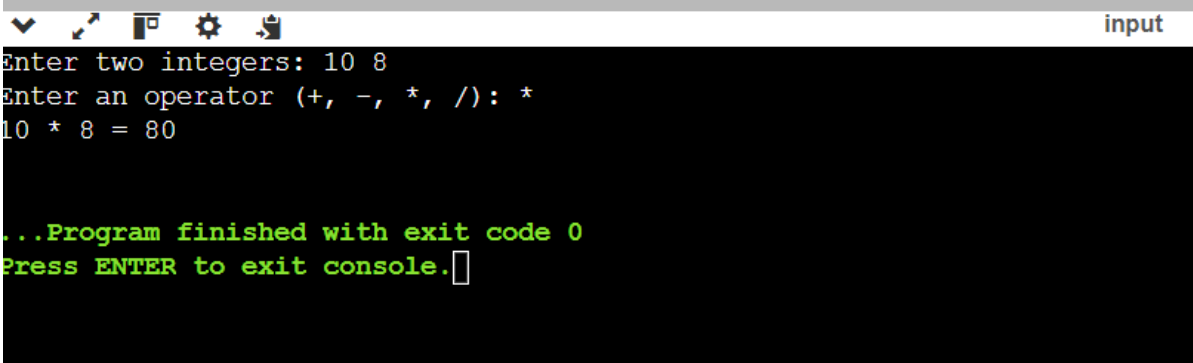
        printf("Invalid operator!\n");

    }

    return 0;

}
```

Output :-

A screenshot of a terminal window with a dark background. The title bar at the top is light gray and contains several icons on the left and the word "input" on the right. The terminal text is as follows:

```
Enter two integers: 10 8
Enter an operator (+, -, *, /): *
10 * 8 = 80

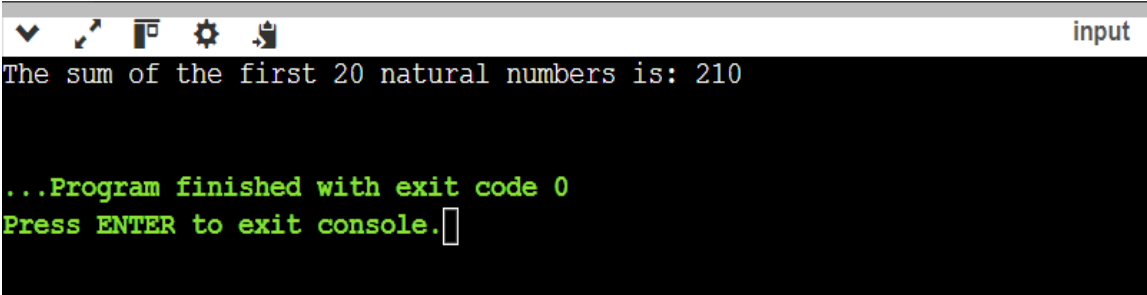
...Program finished with exit code 0
Press ENTER to exit console.
```

2) Write a C program to calculate the sum of the first 20 natural numbers using a recursive function.

Input :-

```
#include <stdio.h>
int sum_of_natural_numbers(int n) {
    if (n == 1) {
        return 1;
    }
    else {
        return n + sum_of_natural_numbers(n - 1);
    }
}
int main() {
    int n = 20;
    int sum = sum_of_natural_numbers(n);
    printf("The sum of the first %d natural numbers is: %d\n", n, sum);
    return 0;
}
```

Output :-

A screenshot of a terminal window with a title bar containing icons for window management and a tab labeled 'input'. The terminal output shows the result of the program execution. The text is displayed in a monospaced font, with the first line in white and the subsequent lines in green.

```
The sum of the first 20 natural numbers is: 210

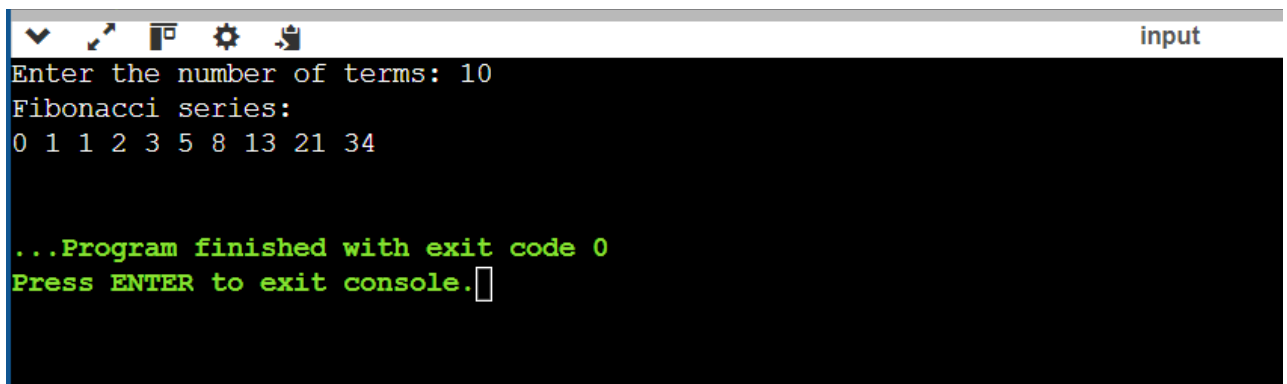
...Program finished with exit code 0
Press ENTER to exit console.
```

3) Write a C program to generate a Fibonacci series using a recursive function.

Input :-

```
#include <stdio.h>
int fibonacci(int n) {
    if (n <= 1) {
        return n;
    }
    else {
        return fibonacci(n - 1) + fibonacci(n - 2);
    }
}
int main() {
    int n, i;
    printf("Enter the number of terms: ");
    scanf("%d", &n);
    printf("Fibonacci series:\n");
    for (i = 0; i < n; i++) {
        printf("%d ", fibonacci(i));
    }
    printf("\n");
    return 0;
}
```

Output :-

A screenshot of a terminal window titled 'input'. The terminal shows the execution of a C program. The user enters '10' for the number of terms. The program outputs the Fibonacci series: '0 1 1 2 3 5 8 13 21 34'. Below the output, it says '...Program finished with exit code 0' and 'Press ENTER to exit console.' with a cursor. The terminal has a dark background with light-colored text.

```
input
Enter the number of terms: 10
Fibonacci series:
0 1 1 2 3 5 8 13 21 34

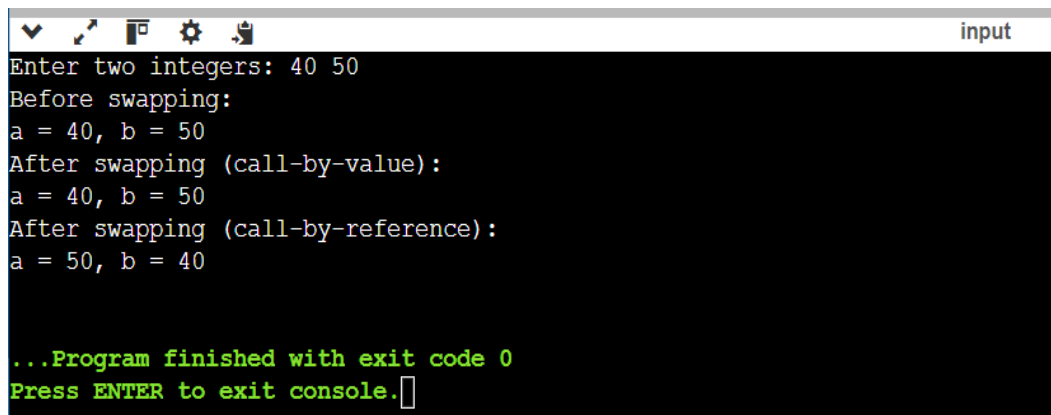
...Program finished with exit code 0
Press ENTER to exit console.
```

4) Write a C program to swap two integers using call-by-value and call-by-reference methods of passing arguments to a function.

Input :-

```
#include <stdio.h>
void swap_by_value(int x, int y) {
    int temp = x;
    x = y;
    y = temp;
}
void swap_by_reference(int *x, int *y) {
    int temp = *x;
    *x = *y;
    *y = temp;
}
int main() {
    int a, b;
    printf("Enter two integers: ");
    scanf("%d %d", &a, &b);
    printf("Before swapping:\n");
    printf("a = %d, b = %d\n", a, b);
    swap_by_value(a, b);
    printf("After swapping (call-by-value):\n");
    printf("a = %d, b = %d\n", a, b);
    swap_by_reference(&a, &b);
    printf("After swapping (call-by-reference):\n");
    printf("a = %d, b = %d\n", a, b);
    return 0;
}
```

Output :-



```
input
Enter two integers: 40 50
Before swapping:
a = 40, b = 50
After swapping (call-by-value):
a = 40, b = 50
After swapping (call-by-reference):
a = 50, b = 40

...Program finished with exit code 0
Press ENTER to exit console.
```

5) Write a C program to find the sum of the digits of the number using a recursive function.

Input :-

```
#include <stdio.h>

int sum_of_digits(int n) {

    if (n == 0) {

        return 0;

    }

    else {

        return n % 10 + sum_of_digits(n / 10);

    }

}

int main() {

    int num;

    printf("Enter a number: ");

    scanf("%d", &num);

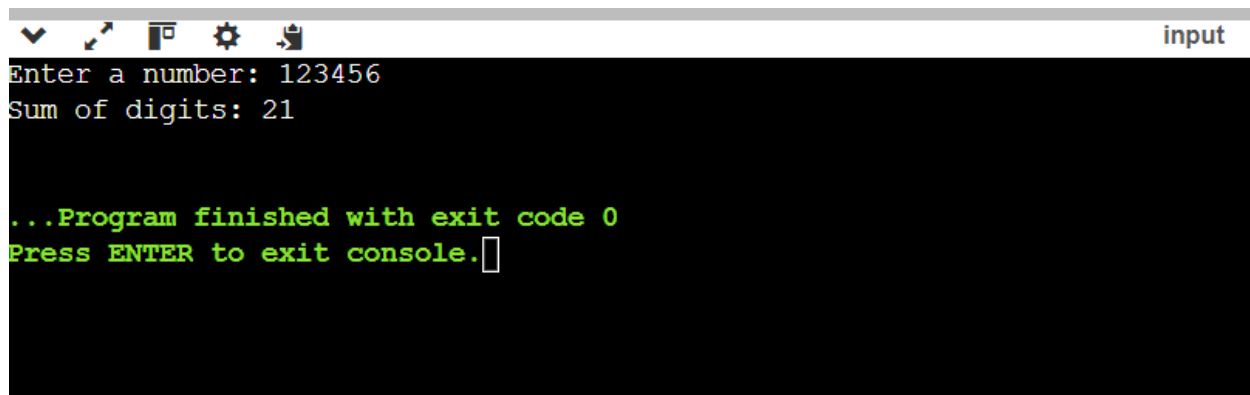
    int sum = sum_of_digits(num);

    printf("Sum of digits: %d\n", sum);

    return 0;

}
```

Output :-

A screenshot of a terminal window with a dark background. The title bar at the top shows standard window controls and the word "input". The terminal text is as follows:

```
Enter a number: 123456
Sum of digits: 21

...Program finished with exit code 0
Press ENTER to exit console.
```

6) Write a C program to read an integer number and print the reverse of that number using recursion.

Input :-

```
#include <stdio.h>
int reverse_number(int n, int reversed) {
    if (n == 0) {
        return reversed;
    }
    else {
        return reverse_number(n / 10, reversed * 10 + n % 10);
    }
}
int main() {
    int num;
    printf("Enter a number: ");
    scanf("%d", &num);
    int reversed = reverse_number(num, 0);
    printf("Reversed number: %d\n", reversed);
    return 0;
}
```

Output :-

A screenshot of a terminal window with a dark background. The title bar at the top shows standard window controls and the word "input". The terminal text is as follows:

```
Enter a number: 12345
Reversed number: 54321

...Program finished with exit code 0
Press ENTER to exit console.
```


7) Using functions, write a C program to find the maximum and minimum between two numbers.

Input :-

```
#include <stdio.h>
```

```
int maximum(int a, int b) {
```

```
    if (a > b) {
```

```
        return a;
```

```
    }
```

```
    else {
```

```
        return b;
```

```
    }
```

```
}
```

```
int minimum(int a, int b) {
```

```
    if (a < b) {
```

```
        return a;
```

```
    }
```

```
    else {
```

```
        return b;
```

```
    }
```

```
}
```

```
int main() {
```

```
    int num1, num2;
```

```
    printf("Enter two numbers: ");
```

```
scanf("%d %d", &num1, &num2);

int max = maximum(num1, num2);

int min = minimum(num1, num2);

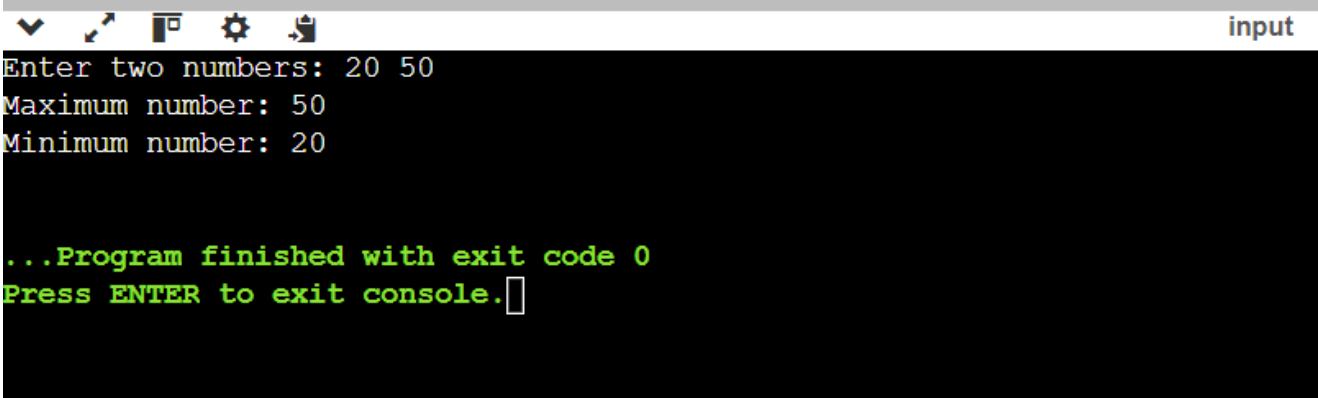
printf("Maximum number: %d\n", max);

printf("Minimum number: %d\n", min);

return 0;

}
```

Output :-

A screenshot of a terminal window with a dark background. The title bar at the top is light gray and contains several icons on the left and the word "input" on the right. The terminal text is as follows:

```
Enter two numbers: 20 50
Maximum number: 50
Minimum number: 20

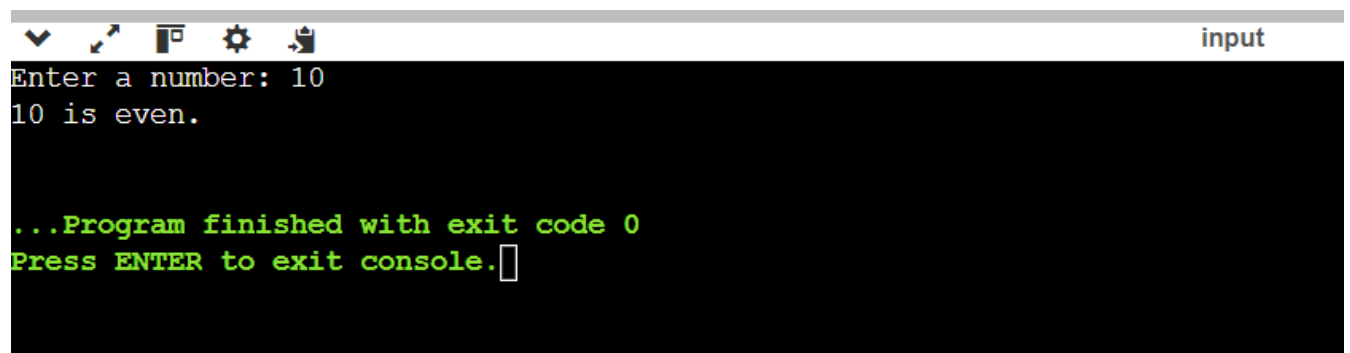
...Program finished with exit code 0
Press ENTER to exit console.
```

8) Write a C program to check whether a number is even or odd using functions.

Input :-

```
#include <stdio.h>
int is_even(int n) {
    if (n % 2 == 0) {
        return 1;
    }
    else {
        return 0;
    }
}
int main() {
    int num;
    printf("Enter a number: ");
    scanf("%d", &num);
    if (is_even(num)) {
        printf("%d is even.\n", num);
    }
    else {
        printf("%d is odd.\n", num);
    }
    return 0;
}
```

Output :-



```
input
Enter a number: 10
10 is even.

...Program finished with exit code 0
Press ENTER to exit console.█
```

9) Write a C program to check whether a number is a prime, Armstrong, or Perfect number using functions.

Input :-

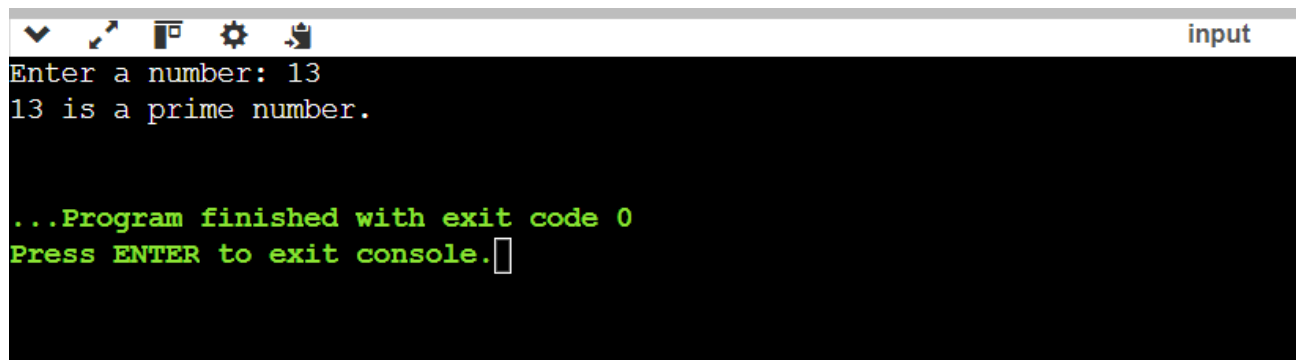
```
#include <stdio.h>
#include <math.h>
int is_prime(int n) {
    if (n <= 1) {
        return 0;
    }
    for (int i = 2; i * i <= n; i++) {
        if (n % i == 0) {
            return 0;
        }
    }
    return 1;
}
int is_armstrong(int n) {
    int original = n, sum = 0, digits = 0;
    while (n != 0) {
        digits++;
        n /= 10;
    }
    n = original;
    while (n != 0) {
        int digit = n % 10;
        sum += pow(digit, digits);
        n /= 10;
    }
    return sum == original;
}
int is_perfect(int n) {
    int sum = 1;
    for (int i = 2; i * i <= n; i++) {
        if (n % i == 0) {
            sum += i + n / i;
        }
    }
    return sum == n;
}
int main() {
    int num;
    printf("Enter a number: ");
```

```

scanf("%d", &num);
if (is_prime(num)) {
    printf("%d is a prime number.\n", num);
}
else if (is_armstrong(num)) {
    printf("%d is an Armstrong number.\n", num);
}
else if (is_perfect(num)) {
    printf("%d is a perfect number.\n", num);
}
else {
    printf("%d is not a prime, Armstrong, or perfect number.\n", num);
}
return 0;
}

```

Output :-



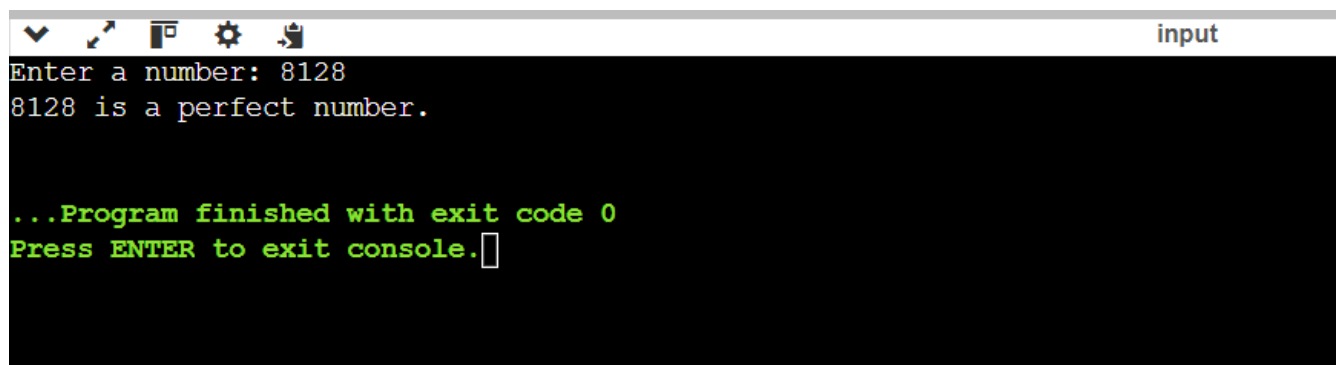
A screenshot of a terminal window with a dark background. The title bar at the top shows standard window controls and the word "input". The terminal text shows the user entering "13" in response to the prompt "Enter a number:". The program outputs "13 is a prime number.". At the bottom, it says "...Program finished with exit code 0" and "Press ENTER to exit console." with a cursor.

```

input
Enter a number: 13
13 is a prime number.

...Program finished with exit code 0
Press ENTER to exit console.

```



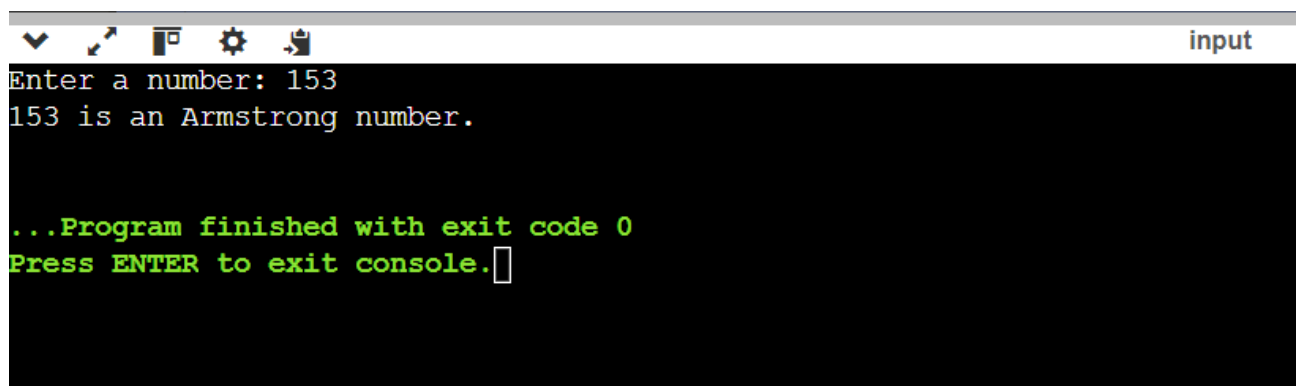
A screenshot of a terminal window with a dark background. The title bar at the top shows standard window controls and the word "input". The terminal text shows the user entering "8128" in response to the prompt "Enter a number:". The program outputs "8128 is a perfect number.". At the bottom, it says "...Program finished with exit code 0" and "Press ENTER to exit console." with a cursor.

```

input
Enter a number: 8128
8128 is a perfect number.

...Program finished with exit code 0
Press ENTER to exit console.

```



A screenshot of a terminal window with a dark background. The title bar at the top shows standard window controls and the word "input". The terminal text shows the user entering "153" in response to the prompt "Enter a number:". The program outputs "153 is an Armstrong number.". At the bottom, it says "...Program finished with exit code 0" and "Press ENTER to exit console." with a cursor.

```

input
Enter a number: 153
153 is an Armstrong number.

...Program finished with exit code 0
Press ENTER to exit console.

```

10) Write a C program to find the power of any number using recursion.

Input :-

```
#include <stdio.h>

int power(int base, int exponent) {

    if (exponent == 0) {

        return 1;

    }

    else {

        return base * power(base, exponent - 1);

    }

}

int main() {

    int base, exponent;

    printf("Enter the base number: ");

    scanf("%d", &base);

    printf("Enter the exponent: ");

    scanf("%d", &exponent);

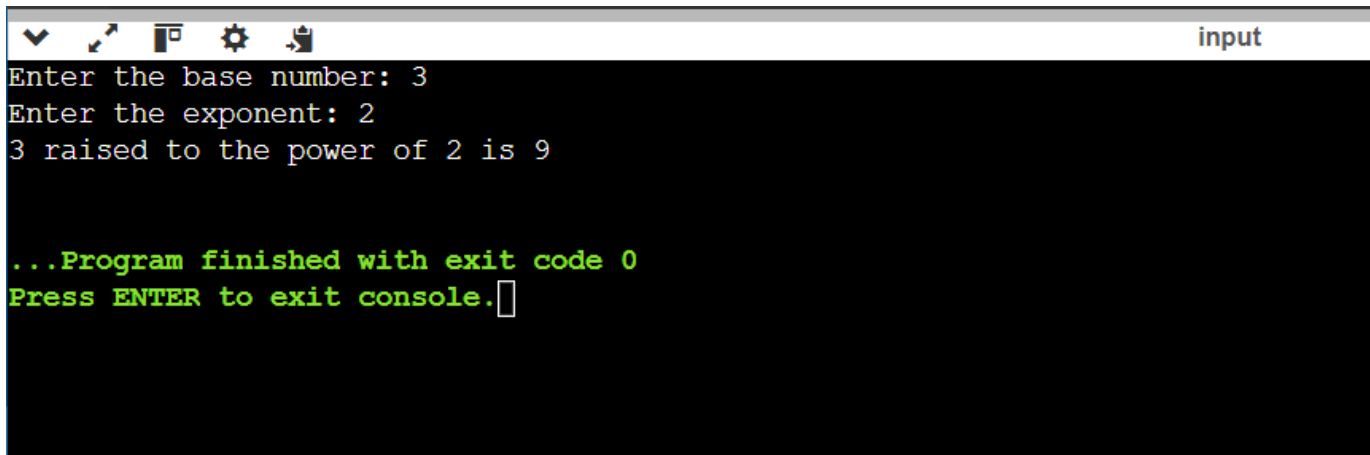
    int result = power(base, exponent);

    printf("%d raised to the power of %d is %d\n", base, exponent, result);

    return 0;

}
```

Output :-



The image shows a terminal window with a title bar containing standard window controls (minimize, maximize, close) and the title 'input'. The terminal has a black background with white text. It displays the following sequence of text: 'Enter the base number: 3', 'Enter the exponent: 2', and '3 raised to the power of 2 is 9'. Below this, in green text, it says '...Program finished with exit code 0' and 'Press ENTER to exit console.' followed by a cursor icon.

```
Enter the base number: 3
Enter the exponent: 2
3 raised to the power of 2 is 9

...Program finished with exit code 0
Press ENTER to exit console.
```